

# Software Requirements Specification (SRS)

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## Algorithmic Trading System

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# 1. Introduction

## 1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to provide a comprehensive and detailed description of the functional and non-functional requirements for the Algorithmic Trading System. This document serves as a blueprint for developers, testers, and stakeholders to ensure a shared understanding of the system's capabilities, constraints, and expectations. It is designed to guide the development process, facilitate testing, and ensure the final product meets all specified requirements.

## 1.2 Scope

The Algorithmic Trading System is a cutting-edge platform designed to automate trading strategies across multiple asset classes, including stocks, ETFs, futures, options, forex, and cryptocurrencies. The system integrates advanced artificial intelligence (AI) and machine learning (ML) technologies to enhance trading decisions, provide real-time analytics, and offer an intuitive user experience. The scope of this document includes:

- Detailed functional requirements for each major component of the system.
- Specifications for external interfaces, including user interfaces, hardware, software, and communication interfaces.
- Non-functional requirements such as performance, security, scalability, usability, and reliability.
- Additional requirements related to legal compliance, auditing, and logging.

This document does not cover:

- Detailed design specifications (covered in the Technical Specification Document).
- Implementation details or code-level requirements.
- Hardware infrastructure provisioning beyond basic requirements.

### 1.3 Definitions, Acronyms, and Abbreviations

- **AI:** Artificial Intelligence
- **ML:** Machine Learning
- **OMS:** Order Management System
- **RAG:** Retrieval-Augmented Generation
- **RBAC:** Role-Based Access Control
- **JWT:** JSON Web Token
- **SSO:** Single Sign-On
- **MFA:** Multi-Factor Authentication
- **PWA:** Progressive Web App
- **WCAG:** Web Content Accessibility Guidelines
- **GDPR:** General Data Protection Regulation
- **HIPAA:** Health Insurance Portability and Accountability Act
- **FIX:** Financial Information Exchange Protocol

## 1.4 References

- Product Requirements Document (PRD)
- Technical Specification Document (TSD)
- API Documentation
- Comprehensive Test Plan
- Features, Phases & Integration Strategy
- Business Requirements Document (BRD)
- Deployment Guide
- System Design Document (SDD)
- Functional Requirements Document (FRD)
- Configuration Management Plan
- Data Flow Document (DFD)

## 1.5 Overview

The remainder of this document is structured as follows:

- **Section 2:** Provides an overall description of the system, including its architecture, user classes, and operating environment.
- **Section 3:** Details the functional requirements for each major component of the system.
- **Section 4:** Specifies the external interface requirements, including user, hardware, software, and communication interfaces.
- **Section 5:** Outlines the non-functional requirements, such as performance, security, and scalability.
- **Section 6:** Covers additional requirements, including legal compliance and auditing.
- **Section 7:** Includes appendices with a glossary and references.

# 2. Overall Description

## 2.1 Product Perspective

The Algorithmic Trading System is a standalone platform that integrates with external brokers and data sources to provide a comprehensive trading solution. It is designed to be modular, scalable, and extensible, allowing for future

enhancements and integrations. The system is built using a microservices architecture, ensuring that each component can be developed, deployed, and scaled independently.

## 2.2 Product Functions

The system provides the following high-level functions:

- **User Management:** Secure registration, authentication, and profile management.
- **Trading Infrastructure:** Multi-asset trading with broker integrations and order management.
- **Data Management:** Real-time and historical data handling with advanced analytics.
- **Strategy Development:** AI-powered tools for creating, backtesting, and optimizing trading strategies.
- **AI Assistant:** Natural language interaction for trading commands and document analysis.
- **User Interface:** Responsive web and mobile access with personalized dashboards.
- **Security and Compliance:** Enterprise-grade features for regulatory adherence and risk management.
- **Options Trading:** Specialized tools for options strategies and risk analysis.

## 2.3 User Classes and Characteristics

The system supports multiple user classes, each with distinct roles and permissions:

- **Admin:** Manages system configurations, user accounts, and audit logs.
- **Trader:** Places and manages orders, views market data, and accesses personal dashboards.
- **Quant:** Develops, backtests, and deploys trading strategies using the Python studio.
- **Risk Manager:** Sets risk parameters, monitors positions, and approves high-risk trades.
- **Compliance Officer:** Accesses audit trails, generates compliance reports, and enforces policies.

## 2.4 Operating Environment

The system is designed to operate in the following environment:

- **Hardware:** Cloud-based infrastructure (e.g., AWS, Azure) with scalable resources.
- **Software:**
  - **Operating System:** Ubuntu 20.04 LTS or later.
  - **Database:** PostgreSQL with pgvector, Qdrant.
  - **Web Server:** Nginx.
  - **Application Server:** Gunicorn (Python), Node.js.
  - **Containers:** Docker, orchestrated with Kubernetes.
- **Network:** High-speed internet connection with low latency for real-time trading.

## 2.5 Design and Implementation Constraints

- The system must use specified open-source technologies (e.g., QuantConnect LEAN, LangChain, Next.js).
- Development must adhere to the 5-phase roadmap outlined in the Features & Phases document.
- The system must comply with financial regulations (e.g., SEC, FINRA) and data protection laws (e.g., GDPR).

## 2.6 Assumptions and Dependencies

- Users have basic knowledge of trading concepts and financial markets.
- External broker APIs and data feeds are reliable and available as per service level agreements (SLAs).
- Sufficient computational resources are available for AI model training and backtesting.
- Third-party libraries and tools (e.g., TA-Lib, pandas-ta) are maintained and updated regularly.

# 3. System Features and Requirements

This section details the functional requirements for each major component of the system. Each requirement is labeled for traceability and includes acceptance

criteria where applicable.

### 3.1 User Management

- **FR-UM-01:** The system shall allow users to register with a unique email and password.
  - *Acceptance Criteria:* Registration form validates email format and password strength (minimum 8 characters, including uppercase, lowercase, number, and special character).
- **FR-UM-02:** The system shall support multi-factor authentication (MFA) using TOTP or authenticator apps.
  - *Acceptance Criteria:* Users can enable MFA and must provide a valid code during login.
- **FR-UM-03:** The system shall integrate Single Sign-On (SSO) with major identity providers.
  - *Acceptance Criteria:* Users can log in using Google or Okta credentials.
- **FR-UM-04:** The system shall provide a secure password reset mechanism with email verification.
  - *Acceptance Criteria:* Password reset links expire after 24 hours and are single-use.
- **FR-UM-05:** The system shall log all authentication events for audit purposes.
  - *Acceptance Criteria:* Audit logs include timestamps, user IDs, and event types (e.g., login, logout).

### 3.2 Core Trading Infrastructure

- **FR-CTI-01:** The system shall integrate with multiple brokers via their APIs.
  - *Details:* Initial integration with Interactive Brokers, expandable to Alpaca and TD Ameritrade.
  - *Acceptance Criteria:* Successful order placement and status retrieval from each broker.
- **FR-CTI-02:** The system shall support trading across asset classes: stocks, ETFs, futures, options, forex, and cryptocurrencies.
  - *Acceptance Criteria:* Users can place orders for any supported asset class with appropriate validation.
- **FR-CTI-03:** The system shall provide an OMS supporting basic and advanced order types.
  - *Details:* Market, Limit, Stop, VWAP, TWAP, etc.

- *Acceptance Criteria:* All order types are executable and tracked in real-time.
- **FR-CTI-04:** The system shall validate orders against user-defined risk limits before submission.
  - *Acceptance Criteria:* Orders exceeding limits are rejected with an error message detailing the violation.
- **FR-CTI-05:** The system shall enable seamless switching between paper and live trading modes.
  - *Acceptance Criteria:* Users can toggle modes without reconfiguring settings, with clear mode indication.

### 3.3 Data Management and Analytics

- **FR-DMA-01:** The system shall integrate with free data sources for paper trading and Interactive Brokers for live trading.
  - *Acceptance Criteria:* Data feeds are accessible and switchable based on trading mode.
- **FR-DMA-02:** The system shall implement a fallback mechanism for data sources.
  - *Details:* Switch to alternative sources if the primary fails.
  - *Acceptance Criteria:* Fallback activates within 5 seconds of primary source failure.
- **FR-DMA-03:** The system shall store historical market data with a minimum depth of 5 years.
  - *Acceptance Criteria:* Data is queryable and exportable in CSV format.
- **FR-DMA-04:** The system shall provide advanced analytics tools, including:
  - Market screener with natural language filters (e.g., "stocks with P/E < 15").
  - AI-powered opportunity detection (e.g., anomaly detection in price movements).
  - Market microstructure analysis (e.g., order book dynamics).
  - *Acceptance Criteria:* Tools generate accurate results within 2 seconds.
- **FR-DMA-05:** The system shall ensure data integrity with checksum validation.
  - *Acceptance Criteria:* Corrupted data is flagged and excluded from processing.

### 3.4 AI-Powered Strategy Development and Backtesting

- **FR-AIS-01:** The system shall provide a no-code strategy builder with a drag-and-drop interface.
  - *User Story:* As a novice trader, I want to create a strategy without coding.
  - *Acceptance Criteria:* Strategies are executable and editable visually.
- **FR-AIS-02:** The system shall include a generative AI tool to convert natural language into strategy code.
  - *Example:* "Buy when 50-day SMA crosses above 200-day SMA."
  - *Acceptance Criteria:* Generated code matches the intent and executes correctly.
- **FR-AIS-03:** The system shall offer a Python-based studio with cloud-hosted Jupyter notebooks.
  - *Acceptance Criteria:* Notebooks support real-time execution and version control.
- **FR-AIS-04:** The system shall provide an enterprise-grade backtesting engine with realistic modeling.
  - *Details:* Slippage, commissions, market impact.
  - *Acceptance Criteria:* Backtest results align with historical data within a 2% margin.
- **FR-AIS-05:** The system shall include a strategy marketplace for sharing and licensing strategies.
  - *Acceptance Criteria:* Users can upload, download, and rate strategies.

### 3.5 Agentic AI Assistant

- **FR-AIA-01:** The system shall provide an AI Assistant for natural language queries.
  - *Examples:* "What's my portfolio value?" or "Place a limit order for 10 shares of AAPL at \$150."
  - *Acceptance Criteria:* Assistant responds accurately within 3 seconds.
- **FR-AIA-02:** The AI Assistant shall execute trading commands based on user instructions.
  - *Acceptance Criteria:* Commands are validated and executed or rejected with feedback.
- **FR-AIA-03:** The AI Assistant shall process user-uploaded documents using Agentic RAG.

- *Acceptance Criteria:* Extracts relevant insights from PDFs within 10 seconds.
- **FR-AIA-04:** The system shall implement a collaborative agent network for task delegation.
  - *Acceptance Criteria:* Complex tasks are broken down and completed by multiple agents.
- **FR-AIA-05:** The AI Assistant shall support continuous learning from user interactions.
  - *Acceptance Criteria:* Response accuracy improves over time, measurable by user feedback.

### 3.6 User Experience and Multi-Platform Accessibility

- **FR-UE-01:** The system shall provide a unified, responsive web interface with personalized dashboards.
  - *Acceptance Criteria:* Dashboards adapt to user preferences and screen sizes.
- **FR-UE-02:** The interface shall integrate institutional-grade charting tools.
  - *Details:* Candlestick charts, technical indicators (e.g., RSI, MACD).
  - *Acceptance Criteria:* Charts render in real-time with <1-second lag.
- **FR-UE-03:** The system shall include a conversational chatbot interface for the AI Assistant.
  - *Acceptance Criteria:* Supports text and voice inputs with 95% accuracy.
- **FR-UE-04:** The system shall support multi-platform access, including PWA capabilities.
  - *Acceptance Criteria:* Installable on iOS and Android devices with offline functionality.
- **FR-UE-05:** The system shall comply with WCAG 2.1 Level AA accessibility standards.
  - *Acceptance Criteria:* Passes automated accessibility audits.

### 3.7 Enterprise-Grade Security, Compliance, and Operations

- **FR-SEC-01:** The system shall implement a Zero-Trust Architecture.
  - *Details:* Continuous authentication and authorization for all requests.
  - *Acceptance Criteria:* Unauthorized access attempts are blocked and logged.

- **FR-SEC-02:** The system shall comply with regulatory standards (e.g., GDPR, HIPAA).
  - *Acceptance Criteria:* Passes compliance audits with no major violations.
- **FR-SEC-03:** The system shall provide automated compliance reporting and audit trails.
  - *Acceptance Criteria:* Reports are generated in PDF format within 5 minutes.
- **FR-SEC-04:** The system shall include a real-time risk management hub with pre-trade checks.
  - *Acceptance Criteria:* Alerts users of risks exceeding thresholds in <1 second.
- **FR-SEC-05:** The system shall support operational monitoring with Prometheus and Grafana.
  - *Acceptance Criteria:* Metrics dashboards update every 5 seconds.

### 3.8 Options Trading

- **FR-OPT-01:** The system shall support trading of basic, complex, and exotic options strategies.
  - *Details:* Spreads, straddles, butterflies, etc.
  - *Acceptance Criteria:* Strategies are executable with real-time pricing.
- **FR-OPT-02:** The system shall provide tools for calculating Greeks and generating payoff diagrams.
  - *Acceptance Criteria:* Calculations are accurate within 0.01% of reference values.
- **FR-OPT-03:** The system shall include a volatility suite with surface constructors.
  - *Acceptance Criteria:* Volatility surfaces render within 3 seconds.
- **FR-OPT-04:** The system shall offer risk management tools specific to options.
  - *Details:* Margin requirements, stress testing.
  - *Acceptance Criteria:* Risk metrics update in real-time.

## 4. External Interface Requirements

### 4.1 User Interfaces

- **UI-01:** The web interface shall be built with Next.js, React, and TypeScript.
  - *Acceptance Criteria:* Interface compiles and runs without errors.
- **UI-02:** The interface shall be responsive and accessible on desktop and mobile devices.
  - *Acceptance Criteria:* Passes responsive design tests across major browsers.
- **UI-03:** The chatbot interface shall support text and voice inputs.
  - *Acceptance Criteria:* Voice recognition achieves 95% accuracy in quiet environments.

## 4.2 Hardware Interfaces

- **HI-01:** The system shall be deployable on standard cloud infrastructure (e.g., AWS EC2 instances).
  - *Acceptance Criteria:* Deploys successfully on specified hardware.
- **HI-02:** The system shall support containerization with Docker for consistent deployment.
  - *Acceptance Criteria:* Containers run without compatibility issues.

## 4.3 Software Interfaces

- **SI-01:** The system shall integrate with broker APIs (e.g., Interactive Brokers, Alpaca).
  - *Acceptance Criteria:* API calls return expected data within 100ms.
- **SI-02:** The system shall use PostgreSQL with pgvector for relational and vector data.
  - *Acceptance Criteria:* Database queries execute within 50ms for typical loads.
- **SI-03:** The system shall use Qdrant for high-performance vector search.
  - *Acceptance Criteria:* Search latency is under 20ms for 1 million vectors.

## 4.4 Communication Interfaces

- **CI-01:** The system shall use HTTPS for secure communication between clients and servers.
  - *Acceptance Criteria:* All traffic is encrypted with TLS 1.3.
- **CI-02:** The system shall support WebSocket for real-time data updates.
  - *Acceptance Criteria:* Updates push to clients within 50ms.

- **CI-03:** The system shall use RESTful APIs for service-to-service communication.
  - *Acceptance Criteria:* APIs conform to OpenAPI 3.0 specifications.

## 5. Other Nonfunctional Requirements

### 5.1 Performance Requirements

- **PR-01:** Order execution latency shall be under 100ms under normal conditions.
  - *Acceptance Criteria:* Measured during load testing with 1,000 concurrent orders.
- **PR-02:** API response times shall be below 100ms for critical operations.
  - *Acceptance Criteria:* 95th percentile latency meets this threshold.
- **PR-03:** The system shall handle at least 10,000 concurrent users.
  - *Acceptance Criteria:* No performance degradation under specified load.

### 5.2 Security Requirements

- **SR-01:** The system shall comply with OWASP Top 10 security standards.
  - *Acceptance Criteria:* Passes OWASP ZAP scans with no high-severity issues.
- **SR-02:** All sensitive data shall be encrypted at rest and in transit.
  - *Acceptance Criteria:* Uses AES-256 encryption and TLS 1.3.
- **SR-03:** The system shall undergo regular security audits and penetration testing.
  - *Acceptance Criteria:* Audits occur quarterly with findings remediated within 30 days.

### 5.3 Scalability Requirements

- **SCR-01:** The system shall support horizontal scaling to handle increased load.
  - *Acceptance Criteria:* Adding nodes reduces latency by at least 20%.
- **SCR-02:** Each microservice shall be independently scalable.
  - *Acceptance Criteria:* Scaling one service does not affect others.

### 5.4 Usability Requirements

- **UR-01:** The interface shall be intuitive with a learning curve of less than 1 hour for basic functions.
  - *Acceptance Criteria:* 90% of new users complete core tasks in usability testing.
- **UR-02:** Comprehensive user guides and documentation shall be provided.
  - *Acceptance Criteria:* Guides cover all features and are accessible online.

## 5.5 Reliability Requirements

- **RR-01:** The system shall achieve 99.99% uptime.
  - *Acceptance Criteria:* Measured over a 3-month period excluding planned maintenance.
- **RR-02:** Automated backups and disaster recovery plans shall be implemented.
  - *Acceptance Criteria:* Recovery time objective (RTO) < 1 hour.

## 6. Other Requirements

### 6.1 Legal and Regulatory Compliance

- **LR-01:** The system shall comply with financial regulations (e.g., SEC, FINRA).
  - *Acceptance Criteria:* Passes regulatory audits with no critical findings.
- **LR-02:** The system shall adhere to data protection laws (e.g., GDPR, CCPA).
  - *Acceptance Criteria:* Includes user consent mechanisms and data deletion options.

### 6.2 Audit and Logging

- **AL-01:** The system shall log all user actions and system events.
  - *Acceptance Criteria:* Logs include timestamps, user IDs, and event details.
- **AL-02:** Audit logs shall be immutable and retained for at least 7 years.
  - *Acceptance Criteria:* Logs are tamper-proof and exportable.

## 7. Appendices

### 7.1 Glossary

- **Agentic RAG:** Retrieval-Augmented Generation with AI agents.
- **Zero-Trust Architecture:** Security model requiring continuous verification.
- **OMS:** Order Management System.
- **Greeks:** Risk metrics for options (e.g., Delta, Gamma).

## 7.2 References

- QuantConnect LEAN Documentation
- LangChain Documentation
- Next.js Documentation
- Interactive Brokers API Documentation

This SRS/SRD provides a detailed and comprehensive specification for the Algorithmic Trading System, ensuring all functional and non-functional requirements are clearly defined for development, testing, and deployment. It serves as a critical guide for all stakeholders involved in the project.